Leveraging Multiple Health Data Sources for Prediction of Hypertension and Heart Disease using Machine Learning Models

Zorica Nikov
zorica.nikov@students.finki.ukim.mk
Faculty of Computer Science and Engineering,
"Ss. Cyril and Methodius" University of Skopje
MK-1000 Skopje, Macedonia

ABSTRACT

Heart disease and hypertension are major global health concerns, affecting millions and often leading to severe complications if not detected early. Predicting these conditions can greatly improve prevention efforts and patient outcomes. In this study, we apply machine learning techniques to analyze correlations between various medical factors, such as age, blood pressure, cholesterol, and lifestyle habits. By combining multiple datasets with similar attributes, we create a unified dataset to identify key features that contribute to predicting these conditions. Our approach focuses on improving early diagnosis and identifying the most significant risk factors.

KEYWORDS

hypertension, heart disease, machine learning, predictive modeling, dataset integration, health predictions [1]

1 INTRODUCTION

Cardiovascular diseases, particularly heart disease and hypertension (HTN), continue to be leading causes of morbidity and mortality worldwide, especially as the global population ages. Conditions such as congestive heart failure (CHF) are prevalent despite ongoing advancements in cardiac treatment. CHF is influenced by multiple risk factors, with hypertension standing out as a major contributor. Although the relationship between HTN and CHF is well-established, the specific mechanisms driving this progression remain unclear [12]. Hypertension is also closely linked to other cardiovascular complications like atrial fibrillation and ventricular arrhythmias. Studies such as the Framingham Heart Study highlight the increased risk of cardiovascular disease in individuals with elevated blood pressure (≥130/85 mmHg), nearly tripling their risk compared to those with normal levels. Additionally, conditions like diabetes and dyslipidemia often coexist with hypertension, compounding the risk of cardiovascular issues [16].

The diagnosis of hypertension is generally made when systolic or diastolic blood pressure readings consistently exceed 120 mmHg or 90 mmHg, respectively. However, identifying and predicting hypertension is challenging due to its dependence on transient environmental factors that can artificially raise blood pressure [14]. Common risk factors for hypertension and heart disease include age, gender, body mass index (BMI), cholesterol levels, stress, smoking habits, and family history [2] [8] [18] [22].

In recent years, machine learning has become an essential tool in medical research, providing powerful methods for diagnosing, detecting, and predicting diseases. Data mining and machine learning techniques have been applied to predict the likelihood of developing heart disease and other conditions, but achieving accurate predictions remains a challenge. This paper aims to explore the application of machine learning algorithms to improve the accuracy of heart disease and hypertension prediction. We utilize a variety of algorithms, including random forest (RF) [17], logistic regression (LR) [9], Support Vector Machines (SVC) [15] and multi-layer perceptron [5] to identify the most significant risk factors and build predictive models.

Given that heart disease affects vital organs, including the kidneys and brain, early detection is critical to prevent severe complications. The World Health Organization (WHO) reported that by 2017, one-third of the global population died from heart disease, making it the leading cause of death in many developing countries. Key risk factors for heart disease include high cholesterol, high blood pressure, diabetes, smoking, obesity, and family history of cardiovascular issues. Common symptoms of heart disease or heart attacks include shortness of breath, chest pain, fatigue, irregular heartbeat, and pain that radiates to the arms, neck, or jaw.

This research will focus on the correlation between medical features, using machine learning models to predict heart disease and hypertension. By merging multiple datasets with overlapping features, we aim to identify the most important factors contributing to these medical conditions. The ultimate goal is to enhance early diagnosis and intervention through accurate prediction models.

2 METHODOLOGY

We will give details about the datasets used, including their purposes, and features. We combined overlapping features from these datasets to create a comprehensive dataset for analysis. Missing values were standardized as -1, and columns with all values as -1 were removed to clean the data during some predictions. The combined dataset was then used to implement various machine learning algorithms to predict heart disease and hypertension. We evaluated the models based on accuracy and other metrics as outlined in the abstract.

2.1 Dataset Selection

The datasets used for the purpose of our paper are as follows:

(1) Heart Failure Prediction Dataset [10] – This dataset consists of 298 clinical records used for predicting the survival of patients with heart failure. It includes various features such as age, sex of the patient, and whether the patient has

[1] hypertension, heart disease, machine learning, predictive modeling, dataset integration, health predictions.

anemia, diabetes, hypertension, along with other important factors like platelet counts in the blood. The dataset also includes biomarkers like serum creatinine and ejection fraction, which are critical for predicting patient outcomes.

(2) Diabetes Health Indicators Dataset [1] – This dataset gathers 253679 responses from Americans regarding health-related risk behaviors and chronic health conditions, collected through the Behavioral Risk Factor Surveillance System survey. It includes features such as whether the patient has diabetes, high blood pressure, cholesterol issues, or any heart disease. Additionally, it records the patient's history of stroke, levels of physical activity, and food consumption patterns.

(3) Heart Stroke Dataset [4] - The dataset gathers information on patients with a history of stroke to analyze its correlation with cardiovascular diseases and the patients' social status. There are 43399 samples.

(4) Identification of Risk Factors for Heart Disease Dataset [21] - The Heart Disease Case Study provides data from 301 heart disease patients, collected by the Hungarian Institute of Cardiology, University Hospital of Zurich, and University Hospital of Basel, resulting in 916 samples. It includes key attributes such as age, sex, chest pain type, blood pressure, cholesterol levels, fasting blood sugar, and maximum heart rate. It helps analyze risk factors, lifestyle influences, and demographic impacts on heart disease.

(5) Hypertension Incidence Dataset [13] – This dataset includes patient information on cardiovascular disease history, family history of such conditions, and lifestyle factors such as smoking and exercise habits in 21612 samples.

2.2 Preprocessing

In the preprocessing phase, we first filtered the relevant features from each dataset, discarding those that were not pertinent to our analysis. For features of the same type but with varying values across different datasets, we standardized them to ensure consistency, converting them to boolean or integer formats based on medical guidelines for attributes such as blood pressure and cholesterol. We then merged all the datasets into a single comprehensive dataset, which includes 12 columns or features:

- Age - integer value indicating the age of the patient at the time of the medical visit.
- Sex - boolean value 0 or 1 indicating female or male respectively.
- Hypertension, or high blood pressure - condition where blood pressure readings are consistently at or above 130/80 mmHg [7].
- Diabetes - chronic medical condition where the body either doesn't produce enough insulin or cannot effectively use the insulin it produces, leading to high blood sugar levels [23].
- Smoking - whether the person has been or is actively smoking.
- Cholesterol - cholesterol levels are measured in milligrams per deciliter (mg/dL). Total cholesterol levels below 200 mg/dL are considered desirable, while levels between 200 and 239 mg/dL are borderline high, and levels 240 mg/dL or higher are high. High-density lipoprotein (HDL) cholesterol levels above 60 mg/dL are protective against heart disease [19].
- Heart diseases - heart diseases encompass various conditions affecting the heart's ability to function properly. For example, coronary artery disease occurs when the arteries supplying blood to the heart become narrow due to plaque buildup. Heart failure involves the heart's inability to pump blood effectively, causing fluid buildup in the lungs and other parts of the body [11].
- BMI - BMI is calculated using the formula: weight (kg) / height^2 (m^2). A BMI below 18.5 is classified as underweight, 18.5-24.9 as normal weight, 25-29.9 as overweight, and 30 or above as obese [3].
- A stroke - medical emergency that occurs when blood flow to a part of the brain is blocked or reduced, leading to brain cell damage [6].
- Physical Activity - boolean value indicating whether the patient exercises or is physically active 2 or more times a week.
- Fasting blood sugar - fasting blood sugar levels are measured in mg/dL after an overnight fast. Normal fasting blood sugar levels are below 100 mg/dL. Prediabetes is indicated by levels between 100 and 125 mg/dL, and diabetes is diagnosed with levels of 126 mg/dL or higher [20].
- Chest Pain Type:
  - Value 1: typical angina
  - Value 2: atypical angina
  - Value 3: non-anginal pain
  - Value 4: asymptomatic

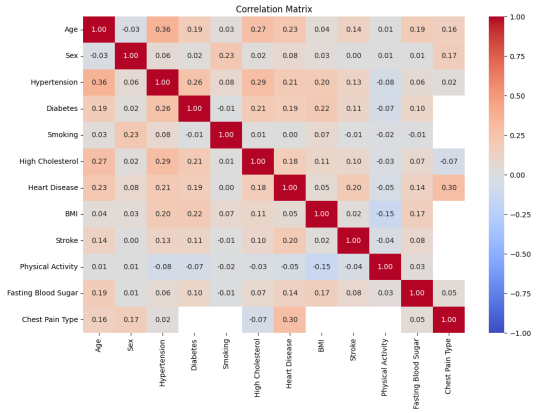

Figure 1: Correlation Matrix of our dataset.

2.3 Visualization

In the visualization phase, several plots were constructed to gain insights into the dataset. Histograms as shown in Figure 2 were created for each feature to show their distributions. A correlation matrix show in Figure 1 was used to display relationships between numerical features, revealing how strongly variables are related to each other.

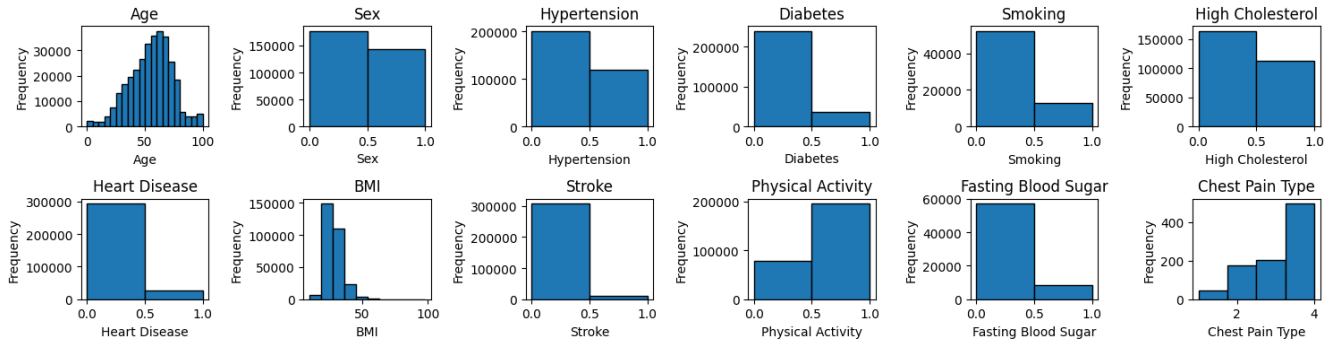

Figure 2: Distribution of features in our dataset.

2.4 Classification

In the modeling phase of our study, we employed various machine learning classifiers to predict medical conditions using the combined dataset. We utilized the following classifiers: Logistic Regression, Random Forest, Support Vector Classifier (SVC) [15], and Neural Network with a multilayer perceptron (MLP) architecture [5] comprising two hidden layers of 400 neurons each.

Initially, the dataset was processed for each target variable, including Hypertension, Heart Disease, Stroke, and Diabetes. For each target, rows with missing values (denoted as -1) in the output column were removed. Additionally, columns where all values were -1 were deleted to ensure the dataset contained only relevant features.

The filtered dataset was then split into training and test sets using a specified split ratio ensuring a consistent random state for reproducibility.

Furthermore, for a more focused analysis on Hypertension, a subset of features relevant to this condition was selected, including Age, BMI, Diabetes, Cholesterol, and Physical Activity. The same preprocessing and modeling steps were applied to this subset to evaluate the classifiers' performance in predicting hypertension specifically. Overall, this comprehensive approach allows for a detailed evaluation of various classifiers in predicting key medical conditions, providing valuable insights into their performance and the importance of different features in the prediction process.

To assess feature importance, we calculated mutual information between each feature and the target variable. A function then evaluated the mutual information score for each feature, highlighting their relevance to predicting the target. This analysis helps identify key features that contribute significantly to predicting conditions such as Hypertension, Heart Disease, Stroke, and Diabetes, providing valuable insights for model refinement and feature selection.

3 RESULTS

The models for predicting hypertension, heart disease, stroke, and diabetes were evaluated using four different classifiers as shown in Figure 3 and Table 1: Logistic Regression, Random Forest, SVC, and Neural Network. Across the conditions, Neural Network consistently provided the highest accuracy, achieving 74.40% for hypertension, 91.88% for heart disease, 96.48% for stroke, and 87.35% for diabetes. SVC and Logistic Regression also performed comparably

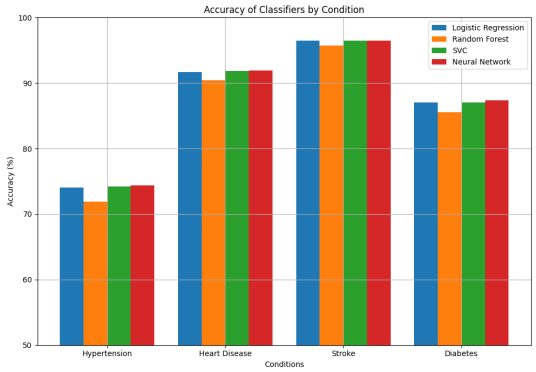

Figure 3: Accuracy of different machine learning models when predicting hypertension, heart disease, stroke and diabetes in our dataset.

Table 1: Model Accuracy for Predicting Hypertension, Heart Disease, Stroke, and Diabetes

Condition	LR	RF	SVC	MLP
Hypertension	74.01%	71.90%	74.20%	**74.40%**
Heart Disease	91.71%	90.47%	91.83%	**91.88%**
Stroke	**96.48%**	95.75%	**96.48%**	**96.48%**
Diabetes	87.06%	85.53%	87.03%	**87.35%**

well, while Random Forest generally achieved slightly lower results, particularly in the prediction of hypertension (71.90%) and diabetes (85.53%).

Mutual information as indicated in Figure 4 was computed to assess the importance of various features in predicting each condition. For hypertension, the most informative features were age (0.0761), high cholesterol (0.0745), and fasting blood sugar (0.0708). In heart disease prediction, chest pain type (0.0835) and fasting blood sugar (0.0717) were the most important features. Stroke prediction was heavily influenced by fasting blood sugar (0.0750) and smoking (0.0405), while diabetes was influenced by smoking (0.0557) and hypertension (0.0547). BMI and age, while relevant, exhibited lower mutual information scores across all conditions.

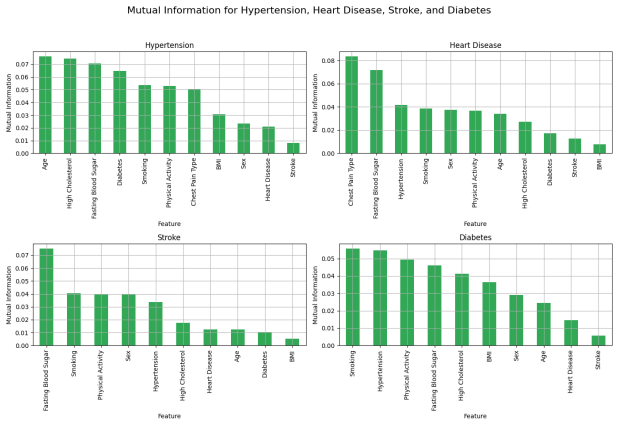

Figure 4: The importance of different features in prediction of the output class.

The visualizations of feature distributions through histograms, correlation matrices, and pair plots helped uncover relationships and trends within the dataset. For instance, fasting blood sugar emerged as a significant indicator across all conditions, reinforcing its role as a critical health metric in disease prediction.

4 CONCLUSION

In this study, we explored the use of machine learning models to predict hypertension, heart disease, stroke, and diabetes by leveraging multiple health datasets. Our results demonstrate that Neural Networks slightly outperform other classifiers across all conditions, indicating their potential as reliable models for disease prediction. Additionally, mutual information analysis revealed key features such as age, fasting blood sugar, and chest pain type as highly influential in the prediction of these conditions.

5 FUTURE WORK

Further refinement of our models could improve prediction accuracy and generalizability, especially by incorporating additional datasets and applying more advanced techniques, like deep learning. Expanding predictions to include more health conditions and using new data sources would further enhance the models. Investigating long-term data and how health conditions evolve over time could provide deeper insights into disease progression, aiding in the development of more effective tools for early detection and prevention of medical conditions. These steps will contribute to improving health outcomes through more accurate predictions.

REFERENCES

[1] BRFSS 2015. 2024. Diabetes Health Indicators Dataset. https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset [Online; accessed 16. Sep. 2024].

[2] Azian Azamimi Abdullah, Zulkarnay Zakaria, and Nur Farahiyah Mohamad. 2011. Design and Development of Fuzzy Expert System for Diagnosis of Hypertension. *ResearchGate* (Feb. 2011), 113–117. https://doi.org/10.1109/ISMS.2011.27

[3] Peymane Adab, Miranda Pallan, and Peter H. Whincup. 2018. Is BMI the best measure of obesity? *BMJ* 360 (March 2018), k1274. https://doi.org/10.1136/bmj.k1274

[4] Lirilkumar Amal. 2024. Heart Stroke. https://www.kaggle.com/datasets/lirilkumaramal/heart-stroke [Online; accessed 16. Sep. 2024].

[5] Eric B. Baum. 1988. On the capabilities of multilayer perceptrons. *J. Complexity* 4, 3 (Sept. 1988), 193–215. https://doi.org/10.1016/0885-064X(88)90020-9

[6] Eran Chemerinski and Robert G. Robinson. 2000. The Neuropsychiatry of Stroke. *Psychosomatics* 41, 1 (Jan. 2000), 5–14. https://doi.org/10.1016/S0033-3182(00)71168-6

[7] Davide Chicco and Giuseppe Jurman. 2020. Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. *BMC Med. Inf. Decis. Making* 20, 1 (Dec. 2020), 1–16. https://doi.org/10.1186/s12911-020-1023-5

[8] Sujit Das, Pijush Kanti Ghosh, and Samarjit Kar. 2013. Hypertension Diagnosis: A Comparative Study using Fuzzy Expert System and Neuro Fuzzy System. *IEEE International Conference on Fuzzy Systems* (July 2013). https://doi.org/10.1109/FUZZ-IEEE.2013.6622434

[9] Stanley Lemeshow Rodney X. Sturdivant David W. Hosmer, Jr. 2013. *Applied Logistic Regression.* Wiley, Hoboken, New Jersey, 2013. https://books.google.mk/books?hl=en&lr=&id=bRoxQBIZRd4C&oi=fnd&pg=PR13&dq=logistics+regression&ots=kM2Qus8Sfc&sig=mwXo6RCvpALsDsP90zc35Vj8G1E&redir_esc=y#v=onepage&q=logistics%20regression&f=false

[10] Giuseppe Jurman Davide Chicco. 2024. Heart Failure Prediction. https://www.kaggle.com/datasets/andrewmvd/heart-failure-clinical-data [Online; accessed 16. Sep. 2024].

[11] PhD Deborah L. Wingard and MD Elizabeth Barrett-Connor. 2022. https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=1269f636576b9a57110ee8d0d8c7db04373551ed [Online; accessed 15. Sep. 2024].

[12] Arun Kannan and Rajesh Janardhanan. 2014. Hypertension as a Risk Factor for Heart Failure. *Curr. Hypertens. Rep.* 16, 7 (July 2014), 1–8. https://doi.org/10.1007/s11906-014-0447-7

[13] Mukhammadjon Kholmirzaev. 2024. Hypertension incidence. https://www.kaggle.com/datasets/kmukhammadjon/hypertension-incidence-in-two-and-half-years/data [Online; accessed 16. Sep. 2024].

[14] Daniel LaFreniere, Farhana Zulkernine, David Barber, and Ken Martin. 2016. Using machine learning to predict hypertension from a clinical dataset. In *2016 IEEE Symposium Series on Computational Intelligence (SSCI)*. IEEE, 06–09. https://doi.org/10.1109/SSCI.2016.7849886

[15] K. W. Lau and Q. H. Wu. 2003. Online training of support vector classifier. *Pattern Recognit.* 36, 8 (Aug. 2003), 1913–1920. https://doi.org/10.1016/S0031-3203(03)00038-4

[16] Gyu Chul Oh and Hyun-Jai Cho. 2020. Blood pressure and heart failure. *Clin. Hypertens.* 26, 1 (Dec. 2020), 1–8. https://doi.org/10.1186/s40885-019-0132-x

[17] Yanjun Qi. 2012. Random Forest for Bioinformatics. In *Ensemble Machine Learning*. Springer, New York, NY, New York, NY, USA, 307–323. https://doi.org/10.1007/978-1-4419-9326-7_11

[18] Rahul Samant and Srikantha Rao. 2013. Evaluation of Artificial Neural Networks in Prediction of Essential Hypertension. *International Journal of Computer Applications* 81, 12 (Nov. 2013), 34–38. https://doi.org/10.5120/14067-2331

[19] Kai Simons and Elina Ikonen. 2000. How Cells Handle Cholesterol. *Science* 290, 5497 (Dec. 2000), 1721–1726. https://doi.org/10.1126/science.290.5497.1721

[20] Anthony Sindoni. 1946. Fasting blood sugar vs. postprandial blood sugar as observed in normal individuals, medical (non-diabetic) patients, and patients with diabetes. *Am. J. Dig. Dis.* 13, 6 (June 1946), 178–192. https://doi.org/10.1007/BF03002761

[21] Swiss Heart Disease Study. 1988. Identification of Risk Factors for Heart Disease. https://www.kaggle.com/datasets/thedevastator/identification-of-risk-factors-for-heart-disease [Online; accessed 16. Sep. 2024].

[22] Mevlut Ture, Imran Kurt, A. Turhan Kurum, and Kazim Ozdamar. 2005. Comparing classification techniques for predicting essential hypertension. *Expert Syst. Appl.* 29, 3 (Oct. 2005), 583–588. https://doi.org/10.1016/j.eswa.2005.04.014

[23] Paul Z. Zimmet, Dianna J. Magliano, William H. Herman, and Jonathan E. Shaw. 2014. Diabetes: a 21st century challenge. *Lancet Diabetes Endocrinol.* 2, 1 (Jan. 2014), 56–64. https://doi.org/10.1016/S2213-8587(13)70112-8